

Problem 1 (Facts about SIS).

- (i) **(Short integer solution exists.)** Let $q(n)$ be prime, $m > n \log q$, $\beta > \sqrt{m}$. Show that for all $A \in \mathbb{Z}_q^{n \times m}$, there exists some short vector $r \in \mathbb{Z}^m$ such that $Ar = \mathbf{0}_n \in \mathbb{Z}_q^n$, $\|r\|_2 \leq \beta$, and $r \neq \mathbf{0}_m$.
- (ii) **(Inhomogeneous SIS is harder than SIS.)** The *Inhomogeneous Short Integer Solution (ISIS)* problem $\text{ISIS}_{n,m,q,\beta}$ with parameters n, m, q, β is the problem that given $A \xleftarrow{\$} \mathbb{Z}_q^{n \times m}$ and $y \xleftarrow{\$} \mathbb{Z}_q^n$, find a “short” vector $r \in \mathbb{Z}^m$ such that $Ar \equiv y \pmod{q}$ and $\|r\|_2 \leq \beta$. As with the SIS problem, for sufficiently large q , $m > n \log q$, and $\beta > \sqrt{m}$, ISIS is believed to be hard, namely, for all PPT algorithms Adv there is a negligible function $\mu(\cdot)$ such that for every $n \in \mathbb{N}$,

$$\Pr \left[A \xleftarrow{\$} \mathbb{Z}_q^{n \times m}, y \xleftarrow{\$} \mathbb{Z}_q^n : r \in \mathbb{Z}^m \leftarrow \text{Adv}(1^n, A, y) \text{ s.t. } Ar \equiv y \pmod{q} \wedge \|r\|_2 \leq \beta \right] \leq \mu(n).$$

Show that there exists some constant c such that for $\hat{m} > m$ and $\hat{\beta} = \beta + c$, $\text{SIS}_{n,\hat{m},q,\hat{\beta}}$ is hard implies $\text{ISIS}_{n,m,q,\beta}$ is hard. In other words, please show that if there is an efficient solver for $\text{ISIS}_{n,m,q,\beta}$ then there is an efficient algorithm that solves $\text{SIS}_{n,\hat{m},q,\hat{\beta}}$.

- (iii) **(SIS is harder than DLWE.)** Let $q = n^{10}$, $\alpha = n^{-5}$, $m > n \log q$, $\beta = O(\sqrt{n \log q})$ and χ be the discrete Gaussian distribution with standard deviation $\alpha \cdot q$. Show that if $\text{DLWE}_{n,m,q,\chi}$ is hard, then $\text{SIS}_{n,m,q,\beta}$ is hard. In other words, show that if there is an efficient solver for $\text{SIS}_{n,m,q,\beta}$ then there is an efficient algorithm that solves $\text{DLWE}_{n,m,q,\chi}$.

Answer (i).

Idea.

$$r \in \{0, 1\}^m \implies \|r\|_2 \leq \sqrt{m} < \beta$$

This follows by the pigeonhole principle. Consider the function $f : \mathbb{Z}^m \rightarrow \mathbb{Z}_q^n$ defined by $f(x) = Ax$. Once restricting f to the domain $\{0, 1\}^m \subseteq \mathbb{Z}^m$, we can see that $|\{0, 1\}^m| = 2^m > 2^{n \log q} = q^n = |\mathbb{Z}_q^n|$, i.e., the size of the new domain is larger than that of the codomain. This implies $f|_{\{0, 1\}^m} : \{0, 1\}^m \rightarrow \mathbb{Z}_q^n$ must be non-injective, meaning that there exist distinct $x_1, x_2 \in \{0, 1\}^m$ such that $f|_{\{0, 1\}^m}(x_1) = f|_{\{0, 1\}^m}(x_2)$. In other words, $Ax_1 = Ax_2$. Therefore, we let $r \triangleq x_1 - x_2 \in \{0, 1\}^m \setminus \{\mathbf{0}_m\}$ (which is because $x_1, x_2 \in \{0, 1\}^m$ and $x_1 \neq x_2$) and then obtain $Ar = A(x_1 - x_2) = Ax_1 - Ax_2 = \mathbf{0}_n$. Lastly, it's easy to verify the second to last requirement for r holds: $\|r\|_2 = \sqrt{r_1^2 + \dots + r_m^2} \leq \sqrt{1^2 + \dots + 1^2} = \sqrt{m} < \beta$, as $r_i \in \{0, 1\}$ for each i . $\otimes$

Answer (ii). Let $c \triangleq 1$.

We prove by contrapositive that for any positive integers $n, \hat{m}, m, \hat{\beta}, \beta$ and prime q such that $\hat{m} > m$ and $\hat{\beta} = \beta + c$, $\text{SIS}_{n,\hat{m},q,\hat{\beta}}$ is hard implies $\text{ISIS}_{n,m,q,\beta}$ is hard.

Let q be a prime and $n, \hat{m}, m, \hat{\beta}, \beta$ be positive integers such that $\hat{m} > m$ and $\hat{\beta} = \beta + c$. Assume that there is an efficient algorithm that solves $\text{ISIS}_{n,m,q,\beta}$, more specifically, that there exists a PPT algorithm Adv such that for all $n \in \mathbb{N}$,

$$\mu(n) \triangleq \Pr \left[A \xleftarrow{\$} \mathbb{Z}_q^{n \times m}, y \xleftarrow{\$} \mathbb{Z}_q^n : r \in \mathbb{Z}^m \leftarrow \text{Adv}(1^n, A, y) \text{ s.t. } Ar \equiv y \pmod{q} \wedge \|r\|_2 \leq \beta \right] \quad (1.1)$$

is a nonnegligible function of n .

By abusing **Adv**, we construct a PPT algorithm $\mathcal{R}$ that solves $\text{SIS}_{n,\hat{m},q,\hat{\beta}}$ as follows.

1: **procedure** $\mathcal{R}(1^n, A \in \mathbb{Z}_q^{n \times \hat{m}})$

2: Parse $n \begin{pmatrix} m & \hat{m} - m \\ A_1 & A_2 \end{pmatrix} := A$, where $A_1 \in \mathbb{Z}_q^{n \times m}$, $A_2 \in \mathbb{Z}_q^{n \times (\hat{m} - m)}$.

3: $r_2 := \begin{pmatrix} 1 & 0 & \cdots & 0 \end{pmatrix}^T \in \mathbb{Z}^{\hat{m} - m}$

4: $r_1 \leftarrow \mathbf{Adv}(1^n, A_1, -A_2 r_2) \in \mathbb{Z}^m$

5: **return** $r := \begin{pmatrix} r_1 \\ r_2 \end{pmatrix} \in \mathbb{Z}^{\hat{m}}$

Clearly $\mathcal{R}$ is a PPT algorithm since **Adv** is a PPT algorithm. Next, observe that given a random block matrix $n \begin{pmatrix} m & \hat{m} - m \\ A_1 & A_2 \end{pmatrix} = A \xleftarrow{\$} \mathbb{Z}_q^{n \times \hat{m}}$ and a constant vector $r_2 \triangleq \begin{pmatrix} 1 & 0 & \cdots & 0 \end{pmatrix}^T \in \mathbb{Z}^{\hat{m} - m}$, since A_1 and A_2 are independent, by a basic property of independence (cf. [Lemma 1.1](#)), A_1 and $-A_2 r_2 = -A_2 \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$ are also independent. Also, since $A_2 = (\alpha_{i,j})_{i \in [n], j \in [\hat{m} - m]}$ for some uniform $\alpha_{i,j} \in \mathbb{Z}_q$, we see that

$$-A_2 r_2 = - \begin{pmatrix} \alpha_{1,1} & \cdots & \alpha_{1,\hat{m}-m} \\ \vdots & \ddots & \vdots \\ \alpha_{n,1} & \cdots & \alpha_{n,\hat{m}-m} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} \alpha_{1,1} \\ \alpha_{2,1} \\ \vdots \\ \alpha_{n,1} \end{pmatrix} \in \mathbb{Z}_q^n$$

is also uniformly random.

Lemma 1.1. Let $X_1, \dots, X_n$ be random variables. If $X_1, \dots, X_n$ are jointly independent, then so are $f_1(X_1), \dots, f_n(X_n)$ for any deterministic functions $f_1, \dots, f_n$ (resp. randomized functions $f_1, \dots, f_n$ where the internal randomness of each f_i is independent of all other random variables).

Therefore, for all $n \in \mathbb{N}$, we denote $r_2 \triangleq \begin{pmatrix} 1 & 0 & \cdots & 0 \end{pmatrix}^T$ and, by the construction of $\mathcal{R}$ and the aforementioned observation, deduce that

$$\begin{aligned} & \Pr \left[A \xleftarrow{\$} \mathbb{Z}_q^{n \times \hat{m}} : r \in \mathbb{Z}^{\hat{m}} \leftarrow \mathcal{R}(1^n, A) \text{ s.t. } r \neq \mathbf{0}_{\hat{m}} \wedge Ar \equiv \mathbf{0}_n \pmod{q} \wedge \|r\|_2 \leq \hat{\beta} \right] \\ &= \Pr \left[A_1 \xleftarrow{\$} \mathbb{Z}_q^{n \times m}, A_2 \xleftarrow{\$} \mathbb{Z}_q^{n \times (\hat{m} - m)} : \right. \\ & \quad \left. r_1 \in \mathbb{Z}^m \leftarrow \mathbf{Adv}(1^n, A_1, -A_2 r_2) \text{ s.t. } \begin{pmatrix} r_1 \\ r_2 \end{pmatrix} \neq \mathbf{0}_{\hat{m}} \wedge \begin{pmatrix} A_1 & A_2 \end{pmatrix} \begin{pmatrix} r_1 \\ r_2 \end{pmatrix} \equiv \mathbf{0}_n \pmod{q} \wedge \left\| \frac{r_1}{r_2} \right\|_2 \leq \hat{\beta} \right] \\ &\stackrel{\text{🍀}}{=} \Pr \left[A_1 \xleftarrow{\$} \mathbb{Z}_q^{n \times m}, A_2 \xleftarrow{\$} \mathbb{Z}_q^{n \times (\hat{m} - m)} : r_1 \in \mathbb{Z}^m \leftarrow \mathbf{Adv}(1^n, A_1, -A_2 r_2) \text{ s.t. } A_1 r_1 \equiv -A_2 r_2 \pmod{q} \wedge \left\| \frac{r_1}{r_2} \right\|_2 \leq \hat{\beta} \right] \\ &\stackrel{\text{🍀}}{\geq} \Pr \left[A_1 \xleftarrow{\$} \mathbb{Z}_q^{n \times m}, A_2 \xleftarrow{\$} \mathbb{Z}_q^{n \times (\hat{m} - m)} : r_1 \in \mathbb{Z}^m \leftarrow \mathbf{Adv}(1^n, A_1, -A_2 r_2) \text{ s.t. } A_1 r_1 \equiv -A_2 r_2 \pmod{q} \wedge \|r_1\|_2 \leq \beta \right] \\ &= \Pr \left[A_1 \xleftarrow{\$} \mathbb{Z}_q^{n \times m}, y \xleftarrow{\$} \mathbb{Z}_q^n : r_1 \in \mathbb{Z}^m \leftarrow \mathbf{Adv}(1^n, A_1, y) \text{ s.t. } A_1 r_1 \equiv y \pmod{q} \wedge \|r_1\|_2 \leq \beta \right] \\ & \quad (\because \text{By the observation above, } y \triangleq -A_2 r_2 \in \mathbb{Z}_q^n \text{ is uniformly random and indepent of } A_1.) \\ &= \mu(n) \quad (\text{by (1.1)}), \end{aligned}$$

which is nonnegligible by assumption. To make it clearer, in the deduction above, the step (♣) follows from the fact that $r_2 \neq \mathbf{0}_{\hat{m}-m}$, which implies that $\begin{pmatrix} r_1 \\ r_2 \end{pmatrix} \neq \mathbf{0}_{\hat{m}}$ always holds, and the fact that

$$\left(\begin{array}{c|c} A_1 & A_2 \end{array} \right) \begin{pmatrix} r_1 \\ r_2 \end{pmatrix} \equiv \mathbf{0}_n \pmod{q} \iff A_1 r_1 + A_2 r_2 \equiv \mathbf{0}_n \pmod{q} \iff A_1 r_1 \equiv -A_2 r_2 \pmod{q};$$

moreover, denoting $r_1 = (\gamma_1 \ \gamma_2 \ \cdots \ \gamma_m)^T$, where $\gamma_i \in \mathbb{Z}$ for each i , the step (♥) follows from a simple observation that

$$\begin{aligned} \|r_1\|_2 = \sqrt{\gamma_1^2 + \dots + \gamma_m^2} \leq \beta &\implies \left\| \frac{r_1}{r_2} \right\|_2 = \sqrt{\gamma_1^2 + \dots + \gamma_m^2 + 1^2} \\ &= \sqrt{\left(\sqrt{\gamma_1^2 + \dots + \gamma_m^2} \right)^2 + 2 \cdot \sqrt{\gamma_1^2 + \dots + \gamma_m^2} \cdot 1 + 1^2} \\ &= \sqrt{\left(\sqrt{\gamma_1^2 + \dots + \gamma_m^2} + 1 \right)^2} \\ &= \sqrt{\gamma_1^2 + \dots + \gamma_m^2} + 1 \\ &= \|r_1\|_2 + 1 \leq \beta + 1 = \beta + c = \hat{\beta}. \end{aligned}$$

Therefore, the PPT algorithm $\mathcal{R}$ indeed solves $\text{SIS}_{n,\hat{m},q,\hat{\beta}}$, which implies that $\text{SIS}_{n,\hat{m},q,\hat{\beta}}$ is not hard. Hence proved. ⊗

Answer (iii). We prove by contrapositive. Assume that there is an efficient algorithm that solves $\text{SIS}_{n,m,q,\beta}$, more specifically, that there exists a PPT algorithm $\mathbf{Adv}$ such that for all $n \in \mathbb{N}$,

$$\mu(n) \triangleq \Pr \left[A \xleftarrow{\$} \mathbb{Z}_q^{n \times m} : r \in \mathbb{Z}^m \leftarrow \mathbf{Adv}(1^n, A) \text{ s.t. } r \neq \mathbf{0}_m \wedge Ar \equiv \mathbf{0}_n \pmod{q} \wedge \|r\|_2 \leq \beta \right] \quad (1.2)$$

is a nonnegligible function of n .

By abusing $\mathbf{Adv}$, we construct a PPT algorithm $\mathcal{R}$ that solves $\text{DLWE}_{n,m,q,\chi}$, as provided in the following experiment $\text{DLWE}_{n,m,q,\chi}$.

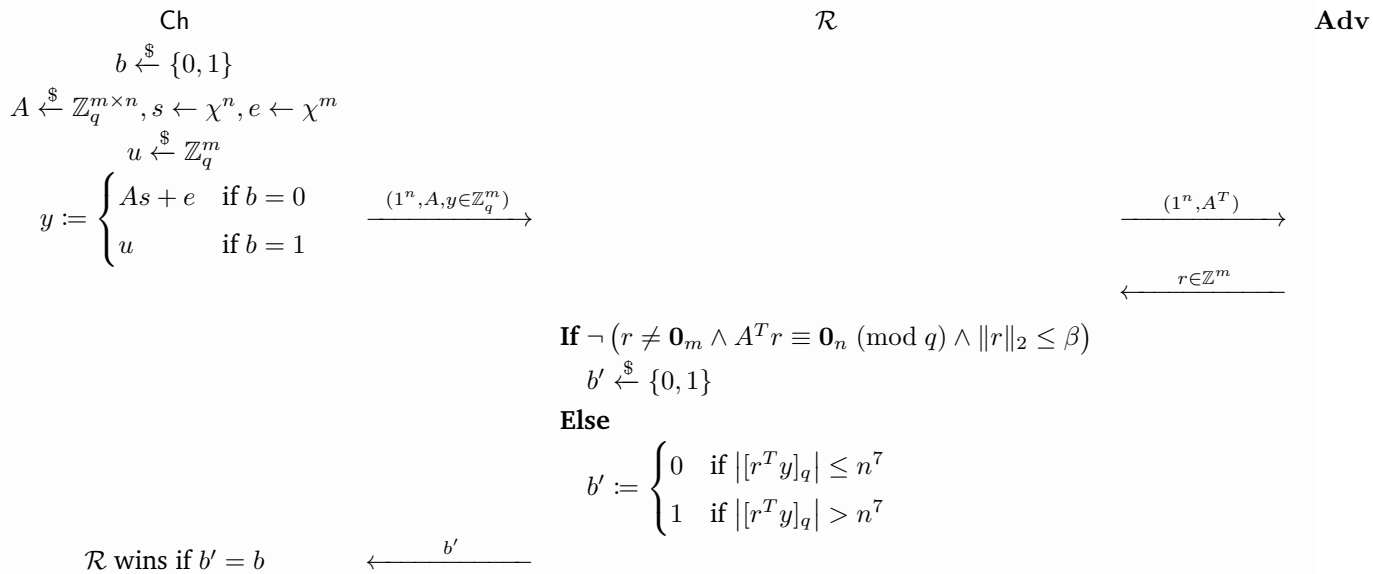


Figure 1: Experiment $\text{DLWE}_{n,m,q,\chi}$

Here we've introduced a notation $[\cdot]_q : \mathbb{Z}_q \rightarrow \mathbb{Z}$ called *centered mod*. For each $x \in \mathbb{Z}_q$, let $[x]_q \in \mathbb{Z}$ denote the *central representative* (aka *centered mod*) of x modulo q , that is, $[x]_q \in [-\frac{q}{2}, \frac{q}{2})$ is such that $[x]_q \bmod q = x \in \mathbb{Z}_q$.

Clearly $\mathcal{R}$ is a PPT algorithm since $\mathbf{Adv}$ is a PPT algorithm.

Denote the event $\text{GOOD} \triangleq r \neq \mathbf{0}_m \wedge A^T r \equiv \mathbf{0}_n \pmod{q} \wedge \|r\|_2 \leq \beta$. Observe that

$$\begin{aligned} \Pr[\text{GOOD}] &= \Pr \left[A \xleftarrow{\$} \mathbb{Z}_q^{m \times n} : r \in \mathbb{Z}^m \leftarrow \mathbf{Adv}(1^n, A^T) \text{ s.t. } r \neq \mathbf{0}_m \wedge A^T r \equiv \mathbf{0}_n \pmod{q} \wedge \|r\|_2 \leq \beta \right] \\ &= \Pr \left[A' \xleftarrow{\$} \mathbb{Z}_q^{n \times m} : r \in \mathbb{Z}^m \leftarrow \mathbf{Adv}(1^n, A') \text{ s.t. } r \neq \mathbf{0}_m \wedge A' r \equiv \mathbf{0}_n \pmod{q} \wedge \|r\|_2 \leq \beta \right] = \mu(n), \end{aligned} \quad (1.3)$$

which is nonnegligible. To analyze the win rate of $\mathcal{R}$ in this experiment, there are 3 cases to consider:

- **Case 1:** $\neg \text{GOOD}$
- **Case 2:** $\text{GOOD} \wedge b = 0$
- **Case 3:** $\text{GOOD} \wedge b = 1$

Clearly conditioning on $\neg \text{GOOD}$ (Case 1), $\mathcal{R}$ has exactly $\frac{1}{2}$ to win since it guesses the answer uniformly at random. For the rest of the cases, we make the following claims.

Claim 1.1 (Case 2). For all $n \in \mathbb{N}$ that are sufficiently large,

$$\Pr \left[|[r^T y]_q| \leq n^7 \mid \text{GOOD} \wedge b = 0 \right] \geq 1 - \frac{1}{n^2}.$$

Proof. As a side note here, elements of $e \in \mathbb{Z}_q^m$ are in $\mathbb{Z}_q$ (as defined by the LWE distribution taught in class), which would make the statistical analysis later a lot trickier. Therefore, for the ease of the proof, we replace $e \leftarrow \chi^m$ with $e' \leftarrow \chi^m$ and redefine $e \triangleq e' \bmod q$, where $e' \in \mathbb{Z}^m$ and $e \in \mathbb{Z}_q^m$, in the experiment.

Conditioning on $\text{GOOD} \wedge b = 0$, we have $y = As + e$, $A^T r = \mathbf{0}_n \in \mathbb{Z}_q^n$, and $\|r\|_2 \leq \beta$. It's easy to see that

$$r^T y = r^T (As + e) = (A^T r)^T s + r^T e = \mathbf{0}_n^T s + r^T e = r^T e = r^T (e' \bmod q) = r^T e' \bmod q \in \mathbb{Z}_q,$$

where all element-wise operations are performed in the field $\mathbb{Z}_q$. Taking centered mod on both sides, we obtain $[r^T y]_q = [r^T e' \bmod q]_q$ and thus $|[r^T y]_q| = |[r^T e' \bmod q]_q|$. Although calculating the probability that $|[r^T y]_q| \leq n^7$ is troublesome, but we may observe that, thanks to the fact that $n^7 < \frac{1}{2}n^{10} = \frac{q}{2}$, if $|r^T e'| \leq n^7$ (without taking centered mod), then surely $|[r^T e' \bmod q]_q| \leq n^7$ (with centered mod). Therefore, we then try to find a lower bound of the probability that $|r^T e'| \leq n^7$.

Observation 1.2. Since

$$\beta = O(\sqrt{n \log q}) = O(\sqrt{10n \log n}) = O(\sqrt{n \log n}) = o(\sqrt{n \cdot n}) = o(n),$$

there exists $N \in \mathbb{N}$ such that $\beta < n$. ■

cont'd. Let $e' = (e'_1 \ e'_2 \ \dots \ e'_m)^T$. **Fixing** $r = (r_1 \ r_2 \ \dots \ r_m)^T \in \mathbb{Z}^m$ (to any value $r^* \in \mathbb{Z}^m$ that is consistent with $\text{GOOD} \wedge b = 0$, i.e., $\Pr[\text{GOOD} \wedge b = 0 \wedge r = r^*] > 0$), we can in fact calculate the variance of $r^T e' \in \mathbb{Z}$ since r is now constant:

$$\begin{aligned} \text{Var}(r^T e' \mid r) &= \text{Var}\left(\sum_{i=1}^m r_i e'_i \mid r\right) = \sum_{i=1}^m \text{Var}(r_i e'_i \mid r) \quad (\because e'_1, \dots, e'_m \text{ are pairwise indep., i.e., } e'_i \perp\!\!\!\perp e'_j \forall i \neq j) \\ &= \sum_{i=1}^m r_i^2 \text{Var}(e'_i \mid r) \\ &= \sum_{i=1}^m r_i^2 \cdot (n^5)^2 \quad (\because e'_i \sim \chi = \Psi_\alpha \implies \sigma_{e'_i} = \alpha \cdot q = n^5) \\ &= n^{10} \|r\|_2^2 \leq n^{10} \beta^2. \end{aligned}$$

By [Observation 1.2](#), for all $n \geq N$, we have

$$\text{Var}(r^T e' \mid r) \leq n^{10} \beta^2 \leq n^{10} n^2 = n^{12}. \quad (1.4)$$

Besides, it's trivial that

$$\mathbb{E}(r^T e' \mid r) = \mathbb{E}\left(\sum_{i=1}^m r_i e'_i \mid r\right) = \sum_{i=1}^m r_i \mathbb{E}(e'_i \mid r) = \sum_{i=1}^m r_i \cdot 0 = 0. \quad (\because e'_i \sim \chi = \Psi_\alpha \implies \mu_{e'_i} = 0)$$

Then, for all $n \geq N$, we apply Chebyshev's inequality to obtain a lower bound of $\Pr[|r^T e'| \leq n^7]$:

$$\Pr[|r^T e'| \leq n^7 \mid r] = \Pr[|r^T e' - \mathbb{E}(r^T e' \mid r)| \leq n^7 \mid r] \geq 1 - \frac{\text{Var}(r^T e' \mid r)}{(n^7)^2} \stackrel{(1.4)}{\geq} 1 - \frac{n^{12}}{n^{14}} = 1 - \frac{1}{n^2}. \quad (1.5)$$

Note again that all probabilities, variances, and expectations above are actually conditioned on $\text{GOOD} \wedge b = 0$ and r is fixed.

Putting it all together, we conclude that for all $n \geq N$,

$$\begin{aligned} &\Pr[|[r^T y]_q| \leq n^7 \mid \text{GOOD} \wedge b = 0] \\ &= \Pr[|[r^T e' \bmod q]_q| \leq n^7 \mid \text{GOOD} \wedge b = 0] \quad (\because [r^T y]_q = [r^T e' \bmod q]_q \text{ by previous observation}) \\ &\geq \Pr[|r^T e'| \leq n^7 \mid \text{GOOD} \wedge b = 0] \quad (\because |r^T e'| \leq n^7 \implies |[r^T e' \bmod q]_q| \leq n^7 \text{ by previous observation}) \\ &= \sum_{r^* \in \mathbb{Z}^m} \Pr[|r^T e'| \leq n^7 \wedge r = r^* \mid \text{GOOD} \wedge b = 0] \\ &= \sum_{r^* \in \mathbb{Z}^m \text{ s.t. } \Pr[\text{GOOD} \wedge b = 0 \wedge r = r^*] > 0} \Pr[r = r^* \mid \text{GOOD} \wedge b = 0] \Pr[|r^T e'| \leq n^7 \mid \text{GOOD} \wedge b = 0 \wedge r = r^*] \\ &\geq \sum_{r^* \in \mathbb{Z}^m \text{ s.t. } \Pr[\text{GOOD} \wedge b = 0 \wedge r = r^*] > 0} \Pr[r = r^* \mid \text{GOOD} \wedge b = 0] \left(1 - \frac{1}{n^2}\right). \quad (\text{by (1.5)}) \\ &= \left(1 - \frac{1}{n^2}\right) \sum_{r^* \in \mathbb{Z}^m} \Pr[r = r^* \mid \text{GOOD} \wedge b = 0] \\ &= \left(1 - \frac{1}{n^2}\right) \Pr[\top \mid \text{GOOD} \wedge b = 0] \\ &= 1 - \frac{1}{n^2}, \end{aligned}$$

as claimed. ■

Claim 1.3 (Case 3). For all $n \in \mathbb{N}$,

$$\Pr \left[|[r^T y]_q| > n^7 \mid \text{GOOD} \wedge b = 1 \right] = 1 - \frac{2n^7 + 1}{n^{10}}.$$

Proof. Conditioning on $\text{GOOD} \wedge b = 1$, we know $y = u \in \mathbb{Z}_q^m$ and $r \in \mathbb{Z}^m$ is nonzero.

Let $u = (u_1 \ u_2 \ \dots \ u_m)^T$. **Fixing** $r = (r_1 \ r_2 \ \dots \ r_m)^T \in \mathbb{Z}^m$ (to any value $r^* \in \mathbb{Z}^m$ that is consistent with $\text{GOOD} \wedge b = 1$, i.e., $\Pr[\text{GOOD} \wedge b = 1 \wedge r = r^*] > 0$), observe that since r is nonzero, there exists some $i^* \in [m]$ such that $r_{i^*} \neq 0$, so we may split the dot product $r^T y = r^T u$ as:

$$r^T u = \sum_{i=0}^m r_i u_i = r_{i^*} u_{i^*} + \sum_{i \in [m] \setminus \{i^*\}} r_i u_i.$$

Since $u_{i^*} \in \mathbb{Z}_q$ is uniform and independent of r_{i^*} , $r_{i^*} u_{i^*} \in \mathbb{Z}_q$ (where multiplication is performed in $\mathbb{Z}_q$) is uniform and thus independent of r_{i^*} . Note that even though r_{i^*} may be dependent on other r_i 's, as $r_{i^*} u_{i^*}$ and r_{i^*} are already independent, together with the fact that $u_1, \dots, u_m$ are jointly independent, we know $r_{i^*} u_{i^*}$ must be independent of other $r_i u_i$'s and thus independent of the rest of the sum $\sum_{i \in [m] \setminus \{i^*\}} r_i u_i$. Again, since $r_{i^*} u_{i^*} \in \mathbb{Z}_q$ is uniform and independent of the rest of the sum $\sum_{i \in [m] \setminus \{i^*\}} r_i u_i \in \mathbb{Z}_q$, the total sum $\sum_{i=1}^m r_i u_i \in \mathbb{Z}_q$ is hence uniform.

As $r^T u = \sum_{i=1}^m r_i u_i \in \mathbb{Z}_q$ is uniform, $[r^T u]_q \in \mathbb{Z}$ must also be uniform over $[-\frac{q-1}{2}, \frac{q-1}{2}]$ (the range of $[\cdot]_q : \mathbb{Z}_q \rightarrow \mathbb{Z}$). Hence, we have

$$\Pr \left[|[r^T u]_q| > n^7 \mid r \right] = 1 - \Pr \left[|[r^T u]_q| \leq n^7 \mid r \right] = 1 - \frac{|[-n^7, n^7]|}{|[-\frac{q-1}{2}, \frac{q-1}{2}]|} = 1 - \frac{2n^7 + 1}{q} = 1 - \frac{2n^7 + 1}{n^{10}}. \quad (1.6)$$

Note again that all probabilities above are actually conditioned on $\text{GOOD} \wedge b = 1$ and r is fixed.

Putting it all together, we conclude that for all $n \in \mathbb{N}$,

$$\begin{aligned} & \Pr \left[|[r^T y]_q| > n^7 \mid \text{GOOD} \wedge b = 1 \right] \\ &= \Pr \left[|[r^T u]_q| > n^7 \mid \text{GOOD} \wedge b = 1 \right] \\ &= \sum_{r^* \in \mathbb{Z}^m} \Pr \left[|[r^T u]_q| > n^7 \wedge r = r^* \mid \text{GOOD} \wedge b = 1 \right] \\ &= \sum_{r^* \in \mathbb{Z}^m \text{ s.t. } \Pr[\text{GOOD} \wedge b = 1 \wedge r = r^*] > 0} \Pr[r = r^* \mid \text{GOOD} \wedge b = 1] \Pr \left[|[r^T u]_q| > n^7 \mid \text{GOOD} \wedge b = 1 \wedge r = r^* \right] \\ &= \sum_{r^* \in \mathbb{Z}^m \text{ s.t. } \Pr[\text{GOOD} \wedge b = 1 \wedge r = r^*] > 0} \Pr[r = r^* \mid \text{GOOD} \wedge b = 1] \left(1 - \frac{2n^7 + 1}{n^{10}} \right) \quad (\text{by (1.6)}) \\ &= \left(1 - \frac{2n^7 + 1}{n^{10}} \right) \sum_{r^* \in \mathbb{Z}^m} \Pr[r = r^* \mid \text{GOOD} \wedge b = 1] \\ &= \left(1 - \frac{2n^7 + 1}{n^{10}} \right) \sum_{r^* \in \mathbb{Z}^m} \Pr[\top \mid \text{GOOD} \wedge b = 1] \\ &= 1 - \frac{2n^7 + 1}{n^{10}}, \end{aligned}$$

as claimed. ■

Lastly, since the input A^T to Adv is independent of b , the output r of Adv must be also independent of b , which can be justified by [Lemma 1.1](#). Thus, (A, r) is independent of b , so GOOD ,

which can be regarded as a binary predicate of A and r , is hence independent of b . This gives us

$$\Pr[b = 0 \mid \text{GOOD}] = \Pr[b = 1 \mid \text{GOOD}] = \Pr[b = 0] = \Pr[b = 1] = \frac{1}{2}. \quad (1.7)$$

Putting it all together, we see that for all $n \in \mathbb{N}$ that are sufficiently large,

$$\begin{aligned} & \Pr[\mathcal{R} \text{ wins DLWE}_{n,m,q,\chi}] \\ &= \Pr[\neg \text{GOOD}] \Pr[b' = b \mid \neg \text{GOOD}] + \Pr[\text{GOOD}] \left(\frac{1}{2} \Pr[b' = 0 \mid \text{GOOD} \wedge b = 0] + \frac{1}{2} \Pr[b' = 1 \mid \text{GOOD} \wedge b = 1] \right) \\ & \quad \text{(by (1.7))} \\ &= (1 - \mu(n)) \cdot \frac{1}{2} + \mu(n) \left(\frac{1}{2} \Pr[|r^T y|_q \leq n^7 \mid \text{GOOD} \wedge b = 0] + \frac{1}{2} \Pr[|r^T y|_q > n^7 \mid \text{GOOD} \wedge b = 1] \right) \\ & \quad \text{(by (1.3))} \\ &\geq (1 - \mu(n)) \cdot \frac{1}{2} + \mu(n) \left(\frac{1}{2} \left(1 - \frac{1}{n^2} \right) + \frac{1}{2} \left(1 - \frac{2n^7 + 1}{n^{10}} \right) \right) \quad \text{(by Claim 1.1 and Claim 1.3)} \\ &= \frac{1}{2} + \frac{1}{2} \mu(n) \left(1 - \frac{1}{n^2} - \frac{2n^7 + 1}{n^{10}} \right). \end{aligned}$$

Note that since $1 - \frac{1}{n^2} - \frac{2(2)^7 + 1}{2^{10}} \leq 1 - \frac{1}{n^2} - \frac{2n^7 + 1}{n^{10}} \leq 1$ and thus

$$1 - \frac{1}{n^2} - \frac{2n^7 + 1}{n^{10}} = \Theta(1) = \frac{1}{n^{\Theta(\log^{-1} n)}} = \frac{1}{n^{O(1)}},$$

so $1 - \frac{1}{n^2} - \frac{2n^7 + 1}{n^{10}}$ is nonnegligible. (Recall that a nonzero function $\varepsilon(\cdot)$ is negligible iff $\varepsilon(n) = \frac{1}{n^{\omega(1)}}$.) Moreover, by assumption we know $\mu(n)$ is nonnegligible. Hence, the advantage $\frac{1}{2} \mu(n) \left(1 - \frac{1}{n^2} - \frac{2n^7 + 1}{n^{10}} \right)$ of $\mathcal{R}$ in experiment $\text{DLWE}_{n,m,q,\chi}$ is nonnegligible.

Therefore, the PPT algorithm $\mathcal{R}$ indeed solves $\text{DLWE}_{n,m,q,\chi}$, which implies that $\text{DLWE}_{n,m,q,\chi}$ is not hard. Hence proved. $\otimes$

Problem 2 (HVZK with abort is sufficient for Fiat-Shamir Signature).

In class, we introduced a lattice-based identification scheme which only satisfies a much weaker security notion that allows the prover to abort, and the aborted transcript is not required to be simulatable. While it may seem odd to aim for such weak notion, this is in fact sufficient to construct a signature scheme from such ID scheme.

(Some definitions)

Show that, if $ID_{\perp}$ is a secure ID scheme with abort that additionally has high first-message min-entropy, then $Sig = FS[ID_{\perp}; H]$ is a secure signature scheme under the random oracle model.

Answer. Let $ID_{\perp} = (KeyGen, P_1, P_2, V)$ be an ID scheme with abort and high first-message min-entropy. Denote by $p(n)$ the abort rate of $ID_{\perp}$ on security parameter $n \in \mathbb{N}$. We prove by contrapositive that $ID_{\perp} = (KeyGen, P_1, P_2, V)$ is secure implies its corresponding Fiat-Shamir signature scheme $Sig \triangleq FS[ID_{\perp}; H] = (KeyGen, Sign, Vrfy)$ is also secure, where H is the random oracle.

Assuming that the signature scheme Sig is not secure, our goal is to show that the ID scheme $ID_{\perp}$ is not secure either. By the security of Sig , there exists a PPT oracle adversary Adv such that

$$\varepsilon(n) \triangleq \Pr \left[Adv^H \text{ wins } Sign^{\text{forge}}(n) \right] \quad (2.1)$$

is nonnegligible (as a function of $n \in \mathbb{N}$), where the EUF-CMA experiment $Sign^{\text{forge}}(n)$ (for each $n \in \mathbb{N}$) is described as follows. Note that in $Sign^{\text{forge}}(n)$, we've expanded the definitions of $Sign$ and $Vrfy$ for convenience, and we've also let $\mathfrak{T}_H$ be a subset of $(\mathcal{M} \times \mathcal{T}) \times \mathcal{C}$, where $\mathcal{M}, \mathcal{T}, \mathcal{C}$ denote the message space, commitment space, and challenge space, respectively. Moreover, for the ease of the proof later, we have the challenger Ch simulate by lazy sampling the random oracle H to which Adv has access.

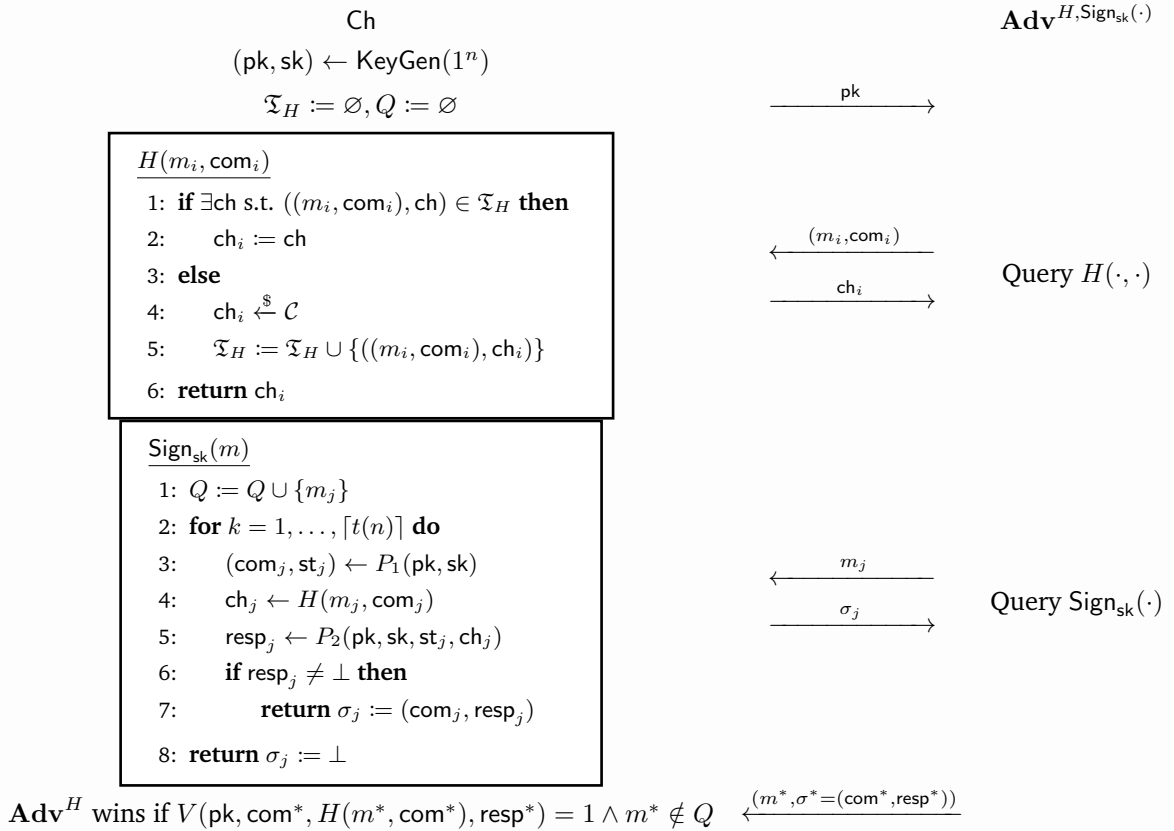


Figure 2: Experiment $Sign^{\text{forge}}(n)$

From this, we construct a PPT adversary $\mathcal{R}$ (without access to the random oracle H) that breaks the security of the underlying ID scheme $\text{ID}_\perp$ with abort. Specifically, we claim that

$$\Pr \left[\mathcal{R} \text{ wins } \text{ID}^{\text{forge}, \perp}(n) \right]$$

is nonnegligible (as a function of n). Note that in order to run/simulate Adv^H , $\mathcal{R}$ must simulate the random oracle H for Adv^H , which is done by lazy sampling in our construction. The construction of $\mathcal{R}$ is given in the following experiment $\text{ID}^{\text{forge}, \perp}(n)$, wherein we denote by $\text{Trans}_{\text{sk}}^\perp$ the transcript oracle and let $\mathfrak{T}_H$ be a subset of $(\mathcal{M} \times \mathcal{T}) \times \mathcal{C}$ for the lazy sampling of H as in experiment $\text{Sign}^{\text{forge}}(n)$.

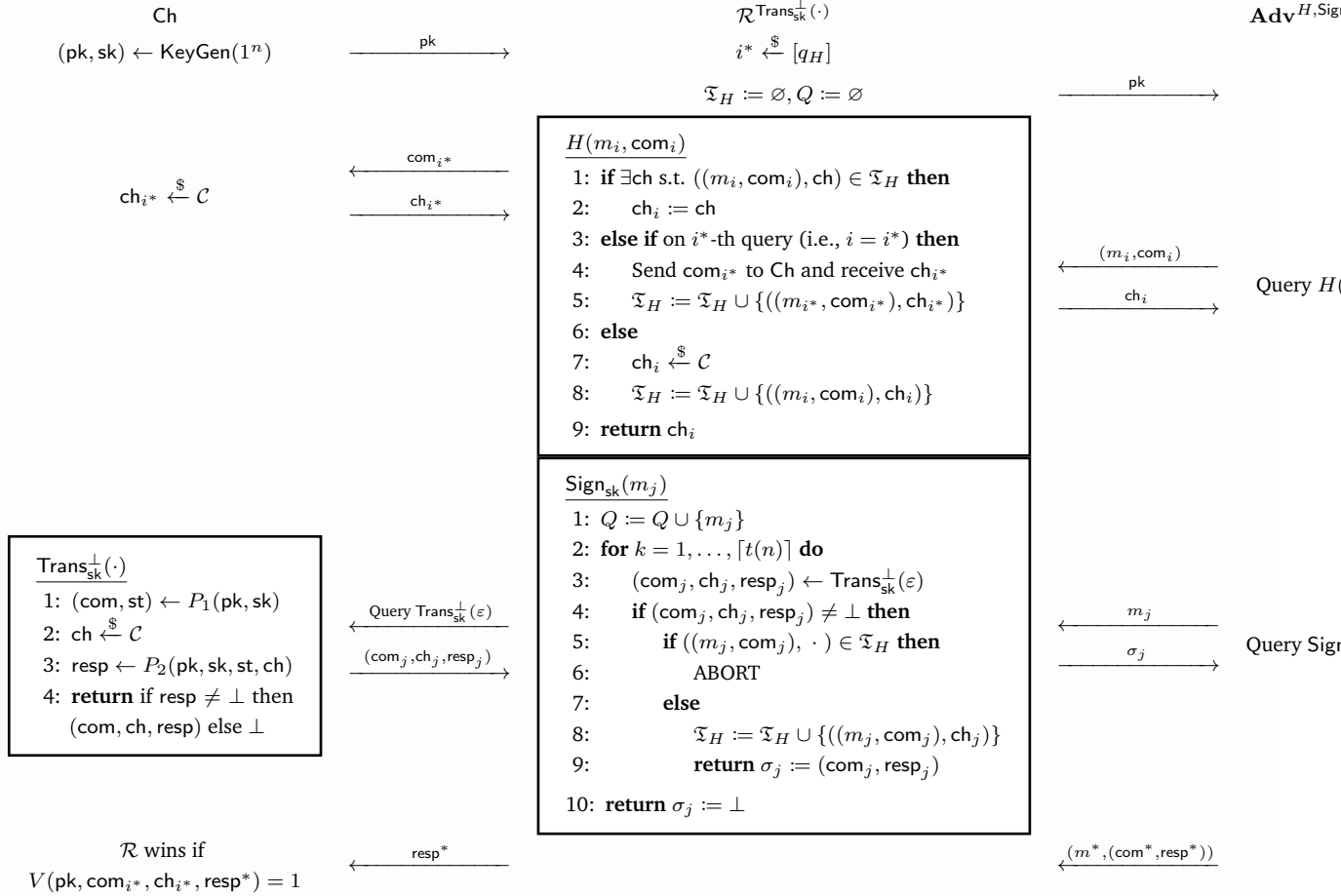


Figure 3: Experiment $\text{ID}^{\text{forge}, \perp}(n)$

Clearly $\mathcal{R}^{\text{Trans}_{\text{sk}}^\perp(\cdot)}$ is a PPT algorithm since $\text{Adv}^{H, \text{Sign}_{\text{sk}}(\cdot)}$ is a PPT algorithm and every oracle call takes (probabilistic) polynomial time.

Let $q_H = O(\text{poly}(n))$ (resp. $q_S = O(\text{poly}(n))$) be an upper bound of the number of issues that $\text{Adv}^{H, \text{Sign}_{\text{sk}}(\cdot)}$ queries to the oracle $H(\cdot, \cdot)$ (resp. $\text{Sign}_{\text{sk}}(\cdot)$). WLOG, assume that all queries made by $\text{Adv}^{H, \text{Sign}_{\text{sk}}(\cdot)}$ are distinct, i.e., that $\text{Adv}^{H, \text{Sign}_{\text{sk}}(\cdot)}$ never makes the same queries to $H(\cdot, \cdot)$ or $\text{Sign}_{\text{sk}}(\cdot)$. Define the event $\text{WIN} \triangleq V(\text{pk}, \text{com}^*, H(m^*, \text{com}^*), \text{resp}^*) = 1 \wedge m^* \notin Q$ in experiment $\text{ID}^{\text{forge}, \perp}(n)$, meaning that $\text{Adv}^{H, \text{Sign}_{\text{sk}}(\cdot)}$ outputs a “correct” answer as if it were playing in its own experiment $\text{Sign}^{\text{forge}}(n)$.

Observe that the distributions of the views of $\text{Adv}^{H, \text{Sign}_{\text{sk}}(\cdot)}$ in experiments $\text{Sign}^{\text{forge}}(n)$ and $\text{ID}^{\text{forge}, \perp}(n)$ are slightly different. This discrepancy in distributions originates from two bad events defined as follows:

- BAD_1 : $\mathcal{R}$ aborts at line 6 in the simulation of oracle $\text{Sign}_{\text{sk}}(\cdot)$. That is, there exists some $j \in [q_S]$ such that at line 5 of $\text{Sign}_{\text{sk}}(\cdot)$, (m_j, com_j) has already been either queried to $H(\cdot, \cdot)$ by $\text{Adv}^{H, \text{Sign}_{\text{sk}}(\cdot)}$ or generated by $\text{Trans}_{\text{sk}}^\perp(\varepsilon)$ during past executions of $\text{Sign}_{\text{sk}}(\cdot)$.
- BAD_2 : $\text{Adv}^{H, \text{Sign}_{\text{sk}}(\cdot)}$ has never queried to $H(\cdot, \cdot)$ (m^*, com^*) , which are returned by $\text{Adv}^{H, \text{Sign}_{\text{sk}}(\cdot)}$ at the end.

It's clear that if BAD_1 and BAD_2 , do not happen, then the probability of WIN in $\text{ID}^{\text{forge}, \perp}(n)$ is equivalent to the probability that $\text{Adv}^{H, \text{Sign}_{\text{sk}}(\cdot)}$ wins in $\text{Sign}^{\text{forge}}(n)$. That is,

$$\Pr[\text{WIN} \mid \neg \text{BAD}_1 \wedge \neg \text{BAD}_2] = \Pr[\text{Adv}^H \text{ wins } \text{Sign}^{\text{forge}}(n) \mid \neg \text{BAD}_2],$$

so

$$\Pr[\text{WIN} \wedge \neg \text{BAD}_1 \wedge \neg \text{BAD}_2] = \Pr[\neg \text{BAD}_1] \Pr[\text{Adv}^H \text{ wins } \wedge \neg \text{BAD}_2 \text{ in } \text{Sign}^{\text{forge}}(n)]. \quad (2.2)$$

Let us find an upper bound of the probability that each bad event occurs.

If BAD_1 occurs in experiment $\text{ID}^{\text{forge}, \perp}(n)$, we know that for some $j \in [q_S]$, during the j -th execution of $\text{Sign}_{\text{sk}}(\cdot)$, (m_j, com_j) matches some past query in the set $\mathfrak{T}_H$, where $(\text{com}_j, \text{ch}_j, \text{resp}_j) \leftarrow \text{Trans}_{\text{sk}}^\perp(\varepsilon)$ is a freshly sampled, valid transcript and $\text{resp}_j \neq \perp$. Recall that $\text{ID}_\perp = (\text{KeyGen}, P_1, P_2, V)$ has a high first-message min-entropy, that is, there exist some $r(n)$ and $\mu(n) = \text{negl}(n)$ such that $2^{-r(n)} = \text{negl}(n)$ and

$$\Pr_{(\text{pk}, \text{sk}) \leftarrow \text{KeyGen}(1^n)} \left[\underbrace{\max_{\widetilde{\text{com}} \in \mathcal{T}} \Pr_{(\text{st}, \text{com}) \leftarrow P_1(\text{pk}, \text{sk})} [\text{com} = \widetilde{\text{com}}]}_{\triangleq \text{GOOD}} \leq 2^{-r(n)} \right] > 1 - \mu(n) \quad (2.3)$$

for all $n \in \mathbb{N}$. Since we know $(\cdot, \text{com}_j) \leftarrow P_1(\text{pk}, \text{sk})$ is fresh, for each $((\widetilde{m}, \widetilde{\text{com}}), \widetilde{\text{ch}}) \in \mathfrak{T}_H$, the probability that (m_j, com_j) collides with $(\widetilde{m}, \widetilde{\text{com}})$ is bounded by:

$$\begin{aligned} & \Pr_{(\text{pk}, \text{sk}) \leftarrow \text{KeyGen}(1^n)} [(m_j, \text{com}_j) = (\widetilde{m}, \widetilde{\text{com}})] \\ &= \Pr_{(\text{pk}, \text{sk}) \leftarrow \text{KeyGen}(1^n)} [\text{GOOD}] \Pr_{(\text{pk}, \text{sk}) \leftarrow \text{KeyGen}(1^n)} [(m_j, \text{com}_j) = (\widetilde{m}, \widetilde{\text{com}}) \mid \text{GOOD}] \\ & \quad + \Pr_{(\text{pk}, \text{sk}) \leftarrow \text{KeyGen}(1^n)} [\neg \text{GOOD}] \Pr_{(\text{pk}, \text{sk}) \leftarrow \text{KeyGen}(1^n)} [(m_j, \text{com}_j) = (\widetilde{m}, \widetilde{\text{com}}) \mid \neg \text{GOOD}] \\ & \stackrel{(2.3)}{\leq} 1 \cdot \Pr_{\substack{(\text{pk}, \text{sk}) \leftarrow \text{KeyGen}(1^n) \\ (\cdot, \text{com}_j) \leftarrow P_1(\text{pk}, \text{sk})}} [\text{com}_j = \widetilde{\text{com}} \mid \text{GOOD}] + \mu(n) \cdot 1 \\ & \stackrel{(2.3)}{\leq} 2^{-r(n)} + \mu(n). \end{aligned} \quad (2.4)$$

Moreover, observe that prior to the j -th query to $\text{Sign}_{\text{sk}}(\cdot)$, (m_j, com_j) , there are at most q_H queries made to $H(\cdot, \cdot)$ and $j - 1$ queries made to $\text{Sign}_{\text{sk}}(\cdot)$. This implies that $\mathfrak{T}_H$ contains at most $q_H + (j - 1)$ query/answer pairs, i.e., $|\mathfrak{T}_H| \leq q_H + (j - 1)$. Therefore, the probability that (m_j, com_j) collides with

ANY $((\widetilde{m}, \widetilde{\text{com}}), \widetilde{\text{ch}}) \in \mathfrak{T}_H$ is bounded by:

$$\begin{aligned}
\Pr \left[\bigvee_{((\widetilde{m}, \widetilde{\text{com}}), \widetilde{\text{ch}}) \in \mathfrak{T}_H} (m_j, \text{com}_j) = (\widetilde{m}, \widetilde{\text{com}}) \right] &\leq \sum_{((\widetilde{m}, \widetilde{\text{com}}), \widetilde{\text{ch}}) \in \mathfrak{T}_H} \Pr [(m_j, \text{com}_j) = (\widetilde{m}, \widetilde{\text{com}})] \quad (\text{by Union Bound}) \\
&\stackrel{(2.4)}{\leq} \sum_{((\widetilde{m}, \widetilde{\text{com}}), \widetilde{\text{ch}}) \in \mathfrak{T}_H} (2^{-r(n)} + \mu(n)) \\
&= (2^{-r(n)} + \mu(n)) \cdot |\mathfrak{T}_H| \\
&\leq (2^{-r(n)} + \mu(n))(q_H + (j-1)). \tag{2.5}
\end{aligned}$$

As we don't actually know $j \in [q_S]$, we simply apply union bound again to sum up the probabilities that $\text{Sign}_{\text{sk}}(\cdot)$ aborts on j -th query for EVERY $j \in [q_S]$:

$$\begin{aligned}
\Pr [\text{BAD}_1] &= \Pr \left[\bigvee_{j=1}^{q_S} \text{Sign}_{\text{sk}}(\cdot) \text{ aborts on } j\text{-th query} \right] \leq \sum_{j=1}^{q_S} \Pr [\text{Sign}_{\text{sk}}(\cdot) \text{ aborts on } j\text{-th query}] \\
&= \sum_{j=1}^{q_S} \Pr \left[\bigvee_{((\widetilde{m}, \widetilde{\text{com}}), \widetilde{\text{ch}}) \in \mathfrak{T}_H} (m_j, \text{com}_j) = (\widetilde{m}, \widetilde{\text{com}}) \right] \\
&\stackrel{(2.5)}{\leq} \sum_{j=1}^{q_S} (2^{-r(n)} + \mu(n))(q_H + (j-1)) \\
&\leq (2^{-r(n)} + \mu(n))(q_H q_S + q_S^2). \tag{2.6}
\end{aligned}$$

On the other hand, in experiment $\text{Sign}^{\text{forge}}(n)$, suppose BAD_2 occurs in $\text{Sign}^{\text{forge}}(n)$, meaning that $\text{Adv}^{H, \text{Sign}_{\text{sk}}(\cdot)}$ has never queried (m^*, com^*) to $H(\cdot, \cdot)$. If we additionally know $m^* \notin Q$, then all query responses that $\text{Adv}^{H, \text{Sign}_{\text{sk}}(\cdot)}$ has received during execution are independent of $H(m^*, \text{com}^*)$, which thus remains uniformly random, albeit possibly set to some value, to $\text{Adv}^{H, \text{Sign}_{\text{sk}}(\cdot)}$. Hence, assuming that conditioning on BAD_2 , for every $\text{pk}, \text{com}, \text{resp}$, there is only one challenge $\text{ch} \in \mathcal{C}$ such that $V(\text{pk}, \text{com}, \text{ch}, \text{resp}) = 1$, this allows us to deduce that:

$$\begin{aligned}
\Pr [\text{Adv}^H \text{ wins } \text{Sign}^{\text{forge}}(n) \mid \text{BAD}_2] &= \Pr [V(\text{pk}, \text{com}^*, H(m^*, \text{com}^*), \text{resp}^*) = 1 \wedge m^* \notin Q \mid \text{BAD}_2] \\
&= \Pr_{\text{ch} \xleftarrow{\$} \mathcal{C}} [V(\text{pk}, \text{com}^*, \text{ch}, \text{resp}^*) = 1 \wedge m^* \notin Q \mid \text{BAD}_2] \\
&\leq \Pr_{\text{ch} \xleftarrow{\$} \mathcal{C}} [V(\text{pk}, \text{com}^*, \text{ch}, \text{resp}^*) = 1 \mid \text{BAD}_2] \\
&= \frac{|\{\text{ch} \in \mathcal{C} \mid V(\text{pk}, \text{com}^*, \text{ch}, \text{resp}^*) = 1\}|}{|\mathcal{C}|} \\
&= \frac{1}{|\mathcal{C}|}. \tag{2.7}
\end{aligned}$$

We further assume $|\mathcal{C}| = \omega(\text{poly}(n))$.

Back in experiment $\text{ID}^{\text{forge}, \perp}(n)$, define the event $\text{GoodGuess} \triangleq ((m_{i^*}, \text{com}_{i^*}) = (m^*, \text{com}^*))$. Conditioning on $\text{WIN} \wedge \neg \text{BAD}_1 \wedge \neg \text{BAD}_2$, from $\neg \text{BAD}_2$ we know $\text{Adv}^{H, \text{Sign}_{\text{sk}}(\cdot)}$ has queried the outputted (m^*, com^*) EXACTLY ONCE, as we've assumed that all queries made by $\text{Adv}^{H, \text{Sign}_{\text{sk}}(\cdot)}$ are distinct. Observe that during each execution of the simulated $H(\cdot, \cdot)$, given the first branch (lines 1-2) is not taken, $\text{Adv}^{H, \text{Sign}_{\text{sk}}(\cdot)}$ cannot distinguish the case $i = i^*$, in which the second branch is taken, from the case $i \neq i^*$, in which the third branch is taken. Therefore, we may see that the uniformly chosen i^* is in fact independent of the view of $\text{Adv}^{H, \text{Sign}_{\text{sk}}(\cdot)}$ and thus its output (m^*, com^*) . This

gives us

$$\begin{aligned}
\Pr[\text{GoodGuess} \mid \text{WIN} \wedge \neg \text{BAD}_1 \wedge \neg \text{BAD}_2] &= \Pr_{i^* \xleftarrow{\$}[q_H]} [(m_{i^*}, \text{com}_{i^*}) = (m^*, \text{com}^*) \mid \text{WIN} \wedge \neg \text{BAD}_1 \wedge \neg \text{BAD}_2] \\
&= \Pr_{i^* \xleftarrow{\$}[q_H]} [(m_{i^*}, \text{com}_{i^*}) = (m_{i^*}, \text{com}_{i^*}) \mid \text{WIN} \wedge \neg \text{BAD}_1 \wedge \neg \text{BAD}_2] \\
&= \Pr_{i^* \xleftarrow{\$}[q_H]} [i^* = i^* \mid \text{WIN} \wedge \neg \text{BAD}_1 \wedge \neg \text{BAD}_2] = \frac{1}{q_H} \quad (2.8)
\end{aligned}$$

since we know $(m^*, \text{com}^*) = (m_{i^*}, \text{com}_{i^*})$ for some $i^* \in [q_H]$ and all queries $(m_1, \text{com}_1), \dots, (m_{q_H}, \text{com}_{q_H})$ are distinct.

Putting it all together, we see that for all $n \in \mathbb{N}$,

$$\begin{aligned}
&\Pr[\mathcal{R} \text{ wins ID}^{\text{forge}, \perp}(n)] \\
&\geq \Pr[\text{WIN} \wedge \neg \text{BAD}_1 \wedge \neg \text{BAD}_2 \wedge \text{GoodGuess}] \\
&= \Pr[\text{WIN} \wedge \neg \text{BAD}_1 \wedge \neg \text{BAD}_2] \Pr[\text{GoodGuess} \mid \text{WIN} \wedge \neg \text{BAD}_1 \wedge \neg \text{BAD}_2] \\
&= \Pr[\text{WIN} \wedge \neg \text{BAD}_1 \wedge \neg \text{BAD}_2] \cdot \frac{1}{q_H} \quad (\text{by (2.8)}) \\
&= \Pr[\neg \text{BAD}_1] \Pr[\mathbf{Adv}^H \text{ wins } \neg \text{BAD}_2 \text{ in Sign}^{\text{forge}}(n)] \cdot \frac{1}{q_H} \quad (\text{by (2.2)}) \\
&= \Pr[\neg \text{BAD}_1] \left(\Pr[\mathbf{Adv}^H \text{ wins Sign}^{\text{forge}}(n)] - \Pr[\mathbf{Adv}^H \text{ wins } \neg \text{BAD}_2 \text{ in Sign}^{\text{forge}}(n)] \right) \cdot \frac{1}{q_H} \\
&\geq \Pr[\neg \text{BAD}_1] \left(\Pr[\mathbf{Adv}^H \text{ wins Sign}^{\text{forge}}(n)] - \Pr[\mathbf{Adv}^H \text{ wins Sign}^{\text{forge}}(n) \mid \text{BAD}_2] \right) \cdot \frac{1}{q_H} \\
&\geq \left(1 - (2^{-r(n)} + \mu(n))(q_H q_S + q_S^2) \right) \left(\varepsilon(n) - \frac{1}{|\mathcal{C}|} \right) \cdot \frac{1}{q_H} \quad (\text{by (2.1), (2.6) and (2.7)}) \\
&= (1 - (\text{negl}(n) + \text{negl}(n))O(\text{poly}(n))) \left(\text{nonnegl}(n) - \frac{1}{\omega(\text{poly}(n))} \right) \cdot \frac{1}{O(\text{poly}(n))} \quad (\text{by assumptions}) \\
&= (1 - \text{negl}(n)) (\text{nonnegl}(n) - \text{negl}(n)) \cdot \text{nonnegl}(n) = \text{nonnegl}(n).
\end{aligned}$$

Hence, by definition, the PPT adversary $\mathcal{R}$ indeed breaks the security of the underlying ID scheme $\text{ID}_\perp$ with abort, meaning that $\text{ID}_\perp$ is not secure. Hence proved. $\circledast$

Problem 3 (Lamport Signature is secure).

In this problem, we will show that we can in fact construct signatures solely from one-way functions. Consider the following security definition of one-time signature: in the one-time signature experiment $\text{Sig} - \text{Forge}_{\text{Adv}, \Pi}^{1\text{-time}}(n)$,

- $\text{KeyGen}(1^n)$ is run to obtain keys (pk, sk) .
- Adversary Adv is given pk and asks a single query m' to its oracle $\text{Sign}_{\text{sk}}(\cdot)$. Adv then outputs (m, σ) .
- The output of the experiment is defined to be 1 if and only if $\text{Vrfy}_{\text{pk}}(m, \sigma) = 1$ and $m \neq m'$.

Signature scheme $\Pi = (\text{KeyGen}, \text{Sign}, \text{Vrfy})$ is *existentially unforgeable under a single-message attack*, or is a *one-time* signature scheme, if for all PPT adversaries Adv , there exists a negligible function $\mu(\cdot)$ such that

$$\Pr \left[\text{Sig} - \text{Forge}_{\text{Adv}, \Pi}^{1\text{-time}}(n) = 1 \right] \leq \mu(n).$$

The Lamport's Signature is described as following: Let $H : \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a OWF. Construct a signature scheme for messages of length $\ell = \ell(n)$ as follows:

- **KeyGen:** On input 1^n , proceed as follows for $i \in [\ell]$:
 - Choose uniform $x_{i,0}, x_{i,1} \in \{0, 1\}^n$.
 - Compute $y_{i,0} := H(x_{i,0})$ and $y_{i,1} := H(x_{i,1})$.

Output the public key $\text{pk} := \{y_{i,0}, y_{i,1}\}_{i=1}^{\ell}$ and the private key $\text{sk} := \{x_{i,0}, x_{i,1}\}_{i=1}^{\ell}$.

- **Sign:** On input a private key sk as above and a message $m \in \{0, 1\}^{\ell}$ with $m = m_1 \cdots m_{\ell}$, output the signature $\sigma := (x_{1,m_1}, \dots, x_{\ell,m_{\ell}})$.
- **Vrfy:** On input a public key pk as above, a message $m \in \{0, 1\}^{\ell}$ with $m = m_1 \cdots m_{\ell}$, and a signature $\sigma = (x_1, \dots, x_{\ell})$, output 1 if and only if $H(x_i) = y_{i,m_i}$ for all $i \in [\ell]$.

Show that the above construction is a one-time signature scheme.

Answer. Let $\Pi = (\text{KeyGen}, \text{Sign}, \text{Vrfy})$ denote the Lamport's signature scheme. To prove the one-timeness of Π , let Adv be a PPT adversary attacking the one-timeness of Π in experiment $\text{Sig} - \text{Forge}_{\text{Adv}, \Pi}^{1\text{-time}}(n)$. Our goal is to prove that $\Pr \left[\text{Sig} - \text{Forge}_{\text{Adv}, \Pi}^{1\text{-time}}(n) = 1 \right] \leq \text{negl}(n)$ for all $n \in \mathbb{N}$ by constructing another adversary (a reduction) $\mathcal{R}$ attacking the one-wayness of $H : \{0, 1\}^n \rightarrow \{0, 1\}^n$ such that $\Pr [\text{OW}_{\mathcal{R}, H}(n) = 1] \leq \text{negl}(n) \implies \Pr \left[\text{Sig} - \text{Forge}_{\text{Adv}, \Pi}^{1\text{-time}}(n) = 1 \right] \leq \text{negl}(n)$, thereby reducing the one-wayness of H to the one-timeness of the signature scheme Π . Note that $\text{OW}_{\mathcal{R}, H}(n)$ denotes the experiment used in the definition of one-wayness, where the function to be tested and the adversary are instantiated with H and $\mathcal{R}$, respectively.

For simplicity, let us split the randomized function Adv into two parts: Adv_1 (randomized) and Adv_2 (deterministic), such that

$$\text{Adv}^{\text{Sign}_{\text{sk}}(\cdot)}(\text{pk}; r) = \text{let } (m', \text{st}) := \text{Adv}_1(\text{pk}; r) \text{ in } \text{Adv}_2(\text{st}, \text{pk}, \text{Sign}_{\text{sk}}(m'))$$

for any key pair (pk, sk) and randomness r . (Note that st denotes the internal state that Adv_1 passes to Adv_2 .)

Now consider the following two games, namely G_0 and G_1 , where G_0 is just the original one-time signature experiment $\text{Sig} - \text{Forge}_{\text{Adv}, \Pi}^{1-\text{time}}(n)$ with an additional $(i^\star, j^\star) \xleftarrow{\$} [\ell] \times \{0, 1\}$, and G_1 is a slightly modified version of G_0 . Note that lines 5-8 are the simulation of the signing oracle $\text{Sign}_{\text{sk}}(\cdot)$, to which Adv will query once throughout each game.

GAMES $G_0^{\text{Adv}}(n), G_1^{\text{Adv}}(n)$	
1: $\text{HIT} := \perp$	▷ Only in G_1
2: $(i^\star, j^\star) \xleftarrow{\$} [\ell] \times \{0, 1\}$	
3: $x_{i,j} \xleftarrow{\$} \{0, 1\}^n, y_{i,j} := H(x_{i,j})$ for each $(i, j) \in [\ell] \times \{0, 1\}$	
4: $\text{sk} := \{x_{i,0}, x_{i,1}\}_{i=1}^\ell, \text{pk} := \{y_{i,0}, y_{i,1}\}_{i=1}^\ell$	
5: $(m' \in \{0, 1\}^\ell, \text{st}) \leftarrow \text{Adv}_1(\text{pk})$	
6: Parse $m'_1 \dots m'_\ell := m'$	
7: if $m'_{i^\star} = j^\star$ then	▷ Only in G_1
8: $\text{HIT} := \top$	▷ Only in G_1
9: $\sigma' := (x_{1,m'_1}, \dots, x_{\ell,m'_\ell}) \in \{0, 1\}^{n \times \ell}$	
10: $(m \in \{0, 1\}^\ell, \sigma \in \{0, 1\}^{n \times \ell}) \leftarrow \text{Adv}_2(\text{st}, \text{pk}, \sigma')$	
11: Parse $m_1 \dots m_\ell := m, (x_1, \dots, x_\ell) := \sigma$	
12: return $\left[\left(\bigwedge_{i=1}^\ell H(x_i) = y_{i,m_i} \right) \wedge m \neq m' \right]$	▷ Only in G_0
13: if $\text{HIT} = \top$ then	▷ Only in G_1
14: return 0	▷ Only in G_1
15: else	▷ Only in G_1
16: return $\left[\left(\bigwedge_{i=1}^\ell H(x_i) = y_{i,m_i} \right) \wedge m_{i^\star} \neq m'_{i^\star} \wedge m_{i^\star} = j^\star \right]$	▷ Only in G_1

Define the set $I \triangleq \{i \in [\ell] \mid m_i \neq m'_i\}$ and the event $\text{WIN} \triangleq \left(\bigwedge_{i=1}^\ell H(x_i) = y_{i,m_i} \right) \wedge m \neq m'$. Conditioning on WIN , since there must be some $i \in [\ell]$ such that $m_i \neq m'_i$, $|I| \geq 1$, meaning that

$$\mathbb{E} \left(\frac{|I|}{\ell} \mid \text{WIN} \right) \geq \frac{1}{\ell}. \quad (3.1)$$

Moreover, since $i^\star$ is independent of all inputs of $\text{Adv}_1, \text{Adv}_2$ and thus independent of their outputs m' and m , $i^\star$ must be independent of $I = \{i \in [\ell] \mid m_i \neq m'_i\}$ as well. Therefore, as $i^\star \in [\ell]$ is uniform, we have

$$\Pr_{i^\star \xleftarrow{\$} [\ell]} [i^\star \in I \mid \text{WIN}] = \mathbb{E} \left(\frac{|I|}{\ell} \mid \text{WIN} \right) \stackrel{(3.1)}{\geq} \frac{1}{\ell}. \quad (3.2)$$

Likewise, since $j^\star$ is independent of all inputs of $\text{Adv}_1, \text{Adv}_2$ and therefore independent of their outputs m' and $(m = m_1 \dots m_\ell, \sigma = (x_1, \dots, x_\ell))$, regardless of the value of $i^\star$, $j^\star$ is surely independent of $m_{i^\star}$. Thus,

$$\Pr_{i^\star \xleftarrow{\$} [\ell], j^\star \xleftarrow{\$} \{0, 1\}} [i^\star \in I \wedge m_{i^\star} = j^\star \mid \text{WIN}] = \Pr_{i^\star \xleftarrow{\$} [\ell]} [i^\star \in I \mid \text{WIN}] \Pr_{j^\star \xleftarrow{\$} \{0, 1\}} [m_{i^\star} = j^\star] \stackrel{(3.2)}{\geq} \frac{1}{\ell} \cdot \frac{1}{2} = \frac{1}{2\ell}. \quad (3.3)$$

(We've used the fact that $m_{i^\star} = j^\star$ and WIN are independent, which can be justified by the aforementioned observation that $j^\star$ is independent of (m, σ) .)

Lastly, it's immediate that

$$\Pr[\text{WIN}] = \Pr \left[\left(\bigwedge_{i=1}^\ell H(x_i) = y_{i,m_i} \right) \wedge m \neq m' \right] = \Pr [G_0^{\text{Adv}}(n) = 1]. \quad (3.4)$$

We can now deduce that for all $n \in \mathbb{N}$,

$$\begin{aligned}
\Pr[G_1^{\text{Adv}}(n) = 1] &= \Pr \left[\left(\bigwedge_{i=1}^{\ell} H(x_i) = y_{i,m_i} \right) \wedge m_{i^\star} \neq m'_{i^\star} \wedge m_{i^\star} = j^\star \wedge m'_{i^\star} \neq j^\star \right] \\
&= \Pr \left[\left(\bigwedge_{i=1}^{\ell} H(x_i) = y_{i,m_i} \right) \wedge m_{i^\star} \neq m'_{i^\star} \wedge m_{i^\star} = j^\star \right] (\because m_{i^\star} \neq m'_{i^\star} \wedge m_{i^\star} = j^\star \implies m'_{i^\star} \neq j^\star) \\
&= \Pr \left[\left(\bigwedge_{i=1}^{\ell} H(x_i) = y_{i,m_i} \right) \wedge m \neq m' \wedge m_{i^\star} \neq m'_{i^\star} \wedge m_{i^\star} = j^\star \right] (\because m_{i^\star} \neq m'_{i^\star} \implies m \neq m') \\
&= \Pr[\text{WIN} \wedge m_{i^\star} \neq m'_{i^\star} \wedge m_{i^\star} = j^\star] \\
&= \Pr[\text{WIN} \wedge i^\star \in I \wedge m_{i^\star} = j^\star] \quad (\text{by definition of } I) \\
&= \Pr[\text{WIN}] \Pr[i^\star \in I \wedge m_{i^\star} = j^\star \mid \text{WIN}] \\
&\geq \frac{1}{2\ell} \Pr[\text{WIN}] \quad (\text{by (3.3)}) \\
&= \frac{1}{2\ell} \Pr[G_0^{\text{Adv}}(n) = 1] \quad (\text{by (3.4)}) \\
&= \frac{1}{2\ell} \Pr[\text{Sig} - \text{Forge}_{\text{Adv}, \Pi}^{1-\text{time}}(n) = 1]. \tag{3.5}
\end{aligned}$$

Having (3.5), we construct a PPT adversary $\mathcal{R}$ attacking the one-wayness of H by running Adv as a subroutine, such that the *view* of Adv in the OW experiment $\text{OW}_{\mathcal{R}, H}(n)$ is identically distributed to that of Adv in game $G_1^{\text{Adv}}(n)$. The construction of $\mathcal{R}$ is given in the OW experiment $\text{OW}_{\mathcal{R}, H}(n)$ (cf. Figure 4).

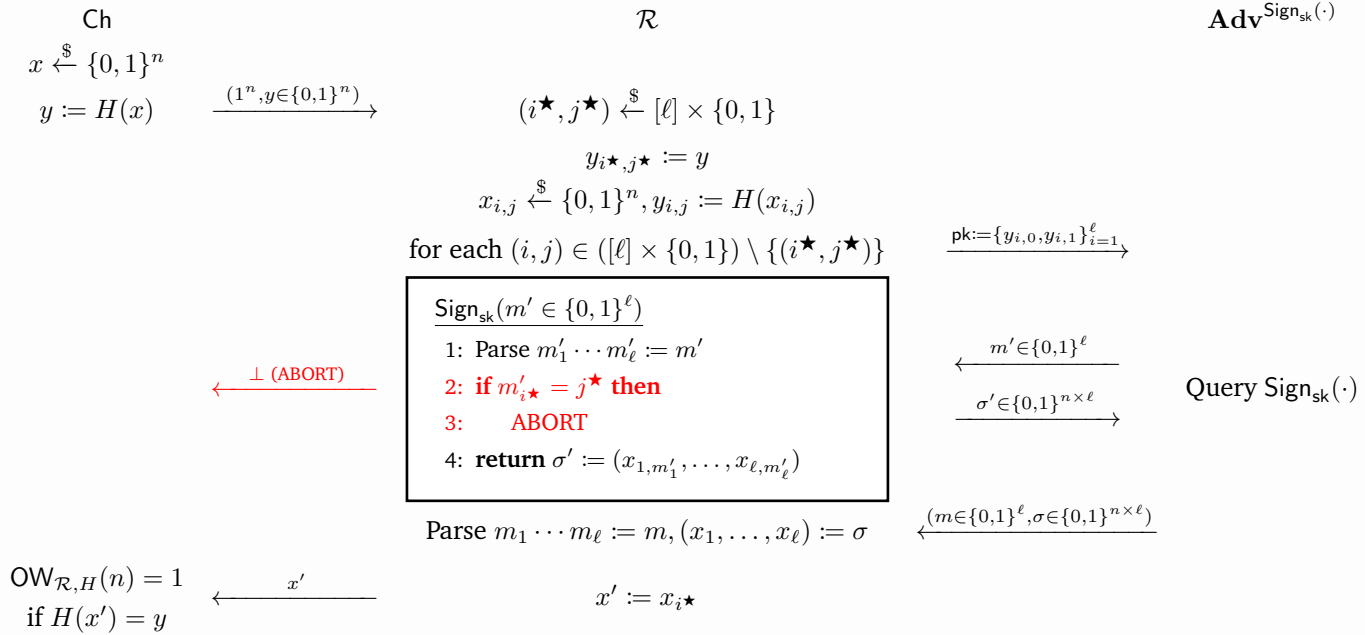


Figure 4: Experiment $\text{OW}_{\mathcal{R}, H}(n)$

Clearly $\mathcal{R}$ is a PPT algorithm since Adv is a PPT algorithm.

It's also clear that the *view* of Adv in the OW experiment $\text{OW}_{\mathcal{R}, H}(n)$ is identically distributed to that of Adv in game $G_1^{\text{Adv}}(n)$, as we previously claimed. Hence, we deduce that for all $n \in \mathbb{N}$, in

experiment $\text{OW}_{\mathcal{R},H}(n)$,

$$\begin{aligned}
\Pr[\text{OW}_{\mathcal{R},H}(n) = 1] &= \Pr[H(x_{i^\star}) = y_{i^\star, j^\star} \wedge \wedge m'_{i^\star} \neq j^\star] \\
&\geq \Pr\left[\left(\bigwedge_{i=1}^{\ell} H(x_i) = y_{i, m_i}\right) \wedge m_{i^\star} \neq m'_{i^\star} \wedge m_{i^\star} = j^\star \wedge m'_{i^\star} \neq j^\star\right] \\
&= \Pr[G_1^{\text{Adv}}(n) = 1] \quad (\because \text{The views of } \text{Adv} \text{ in } \text{OW}_{\mathcal{R},H}(n) \text{ and } G_1^{\text{Adv}}(n) \text{ are iden. dist.}) \\
&\geq \frac{1}{2\ell} \Pr[\text{Sig} - \text{Forge}_{\text{Adv}, \Pi}^{1\text{-time}}(n) = 1]. \quad (\text{by (3.5)}) \tag{3.6}
\end{aligned}$$

Since H is assumed to be one-way and $\mathcal{R}$ is a PPT adversary attacking the one-wayness of H , by definition, there exists a negligible function $\varepsilon(\cdot)$ such that for all $n \in \mathbb{N}$,

$$\Pr[\text{OW}_{\mathcal{R},H}(n) = 1] \leq \varepsilon(n). \tag{3.7}$$

Assume $\ell = \ell(n) = O(\text{poly}(n))$, which is reasonable because the signature scheme $\Pi = (\text{KeyGen}, \text{Sign}, \text{Vrfy})$ has to be efficiently computable (in particular, polynomial-time computable) to be actually useful. Combining (3.6) and (3.7), we finally conclude that for all $n \in \mathbb{N}$,

$$\Pr[\text{Sig} - \text{Forge}_{\text{Adv}, \Pi}^{1\text{-time}}(n) = 1] \leq 2\ell \Pr[\text{OW}_{\mathcal{R},H}(n) = 1] \leq 2\ell\varepsilon(n) = 2O(\text{poly}(n))\text{negl}(n),$$

which is still negligible. As the choice of the PPT adversary Adv is arbitrary, by definition, the signature scheme $\Pi = (\text{KeyGen}, \text{Sign}, \text{Vrfy})$ is a one-time signature scheme. This completes our proof.

⊗

Reference

- P1 (i) [Lecture notes by João Ribeiro](#)
- (ii) [Stack Exchange - The equivalence of SIS and ISIS\(Inhomogeneous SIS\)](#)
- (iii) [Basic Lattice Cryptography: The concepts behind Kyber \(ML-KEM\) and Dilithium \(ML-DSA\) by Vadim Lyubashevsky](#)
- P2 [TAMC 2026 Week 11 Digital Signature Slides](#)
- P3 (None)